When do agent fleets improve web development?

Lecture 7 · Compare

Parallel generation helps only when decomposition, integration, and review can keep up.

A fleet changes the coordination problem

Adding agents can increase:

- work attempted in parallel;
- context boundaries and handoffs;
- duplicated exploration;
- integration conflicts;
- review and validation demand;
- the authority exposed to automated work.

The objective is accepted outcomes per constrained human attention—not maximum activity.

Decompose around contracts and evidence

Good delegated tasks have:

- one observable outcome;
- a bounded change surface;
- explicit inputs and dependencies;
- an acceptance check;
- a named integrator or owner;
- a stop or escalation condition.

Splitting by files alone can distribute code while leaving the real design decision shared and unresolved.

Roles should describe authority, not personalities

Role	Typical authority	Required evidence
Explorer	Propose alternatives	Options, assumptions, unknowns
Implementer	Change a bounded surface	Diff and focused checks
Reviewer	Evaluate against declared criteria	Findings and verdict
Integrator	Resolve contracts and combine accepted work	Integrated checks and known limits
Operator	Exercise deployment or recovery	Run evidence and rollback state

The task graph—not agent count—sets the speed limit

Map:

- independent branches;
- sequential prerequisites;
- shared resources and contracts;
- integration gates;
- human decisions on the critical path.

Parallel tasks that all wait for one ambiguous interface do not create parallel progress.

Amdahl’s Law bounds ideal speedup

If 80% of the work can be parallelized, the sequential 20% remains.

S(N) = 1 / ((1 - 0.8) + 0.8 / N)

With four workers, ideal speedup is **2.5×**, not 4×.

As N grows, maximum ideal speedup approaches **5×**—before coordination, integration, and review overhead.

Integration is part of the fleet design

Before parallel work begins, define:

- shared contracts and invariants;
- branch or workspace isolation;
- validation gates for delegated tasks;
- how failures and conflicts are surfaced;
- who can accept the integrated revision;
- which external actions require human approval.

The fleet is not complete until its outputs converge into one reviewable artifact.

Launch Compare with a falsifiable fleet plan

Produce:

- a task graph and critical path;
- role, context, tool, and permission boundaries;
- integration and validation gates;
- a single-agent counterfactual;
- one predicted benefit and one predicted coordination cost;
- a trace that can later explain a failed or corrected task.

Question to carry forward: Which scaling response is justified by the application evidence?